

## FLEXHAND · PROBLEM STATEMENT

# FlexHand problem statement

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## DOCUMENT CONTROL

|             |                                       |         |                |
|-------------|---------------------------------------|---------|----------------|
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| Author      | Amish Chadha                          | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept         |         |                |
| Repository  | github.com/BoujeeEnjinia1701/flexhand |         |                |
| Source      | docs/01-problem.md @ uncommitted      |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                                                 |
|---------|------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.6     | 2026-10-02 | Amish Chadha | Open question on the first clinical partner answered by the decision of 2026-10-02 (FXH-DEC-001)                                                       |
| 0.5     | 2026-10-01 | Amish Chadha | Parts cost stated as a value-engineering target                                                                                                        |
| 0.4     | 2026-09-25 | Amish Chadha | Recommendations accepted by Amish (DDR-002); first severity band set to mild flexor tone                                                               |
| 0.3     | 2026-09-25 | Amish Chadha | Record decisions of FXH-DDR-001 (passive motion only, design load); complete the passive-mobilization citation; severity band finding from FXH-CAL-001 |
| 0.2     | 2026-09-25 | Amish Chadha | Populate to TRL 2 (problem, users, context, constraints, prior work)                                                                                   |
| 0.1     | 2026-09-24 | Amish Chadha | Initial scaffold                                                                                                                                       |

## FlexHand problem statement

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Recovery of hand function after stroke seems to depend on a large number of movement repetitions, but a typical therapy session delivers only a few dozen, and the rest of the day the affected hand often stays still and curled into flexion. FlexHand is a low-cost, open, wearable device that moves the fingers through hundreds of controlled flexion and extension cycles a day at home, within limits a therapist sets.

### ■ The problem

Hand impairment is common and persistent after stroke. In a cohort of people who had a flaccid arm soon after stroke, only about 38 % had regained some dexterity at six months ([Kwakkel et al., \*Stroke\*, 2003](#)). Difficulty opening the hand is a large part of this impairment: finger extension deficits after stroke arise from both neural factors and changed muscle mechanics ([Kamper and Rymer, \*Muscle & Nerve\*, 2003](#)), and finger extension is typically affected more than flexion ([Soft pneumatic actuators for pushing fingers into extension, \*JNER\*, 2024](#)).

Therapy dose is small. Lang and colleagues observed 312 physical and occupational therapy sessions and found an average of 32 repetitions of upper-limb functional movement per session, against the 400 to 600 repetitions per session used in animal models of recovery ([Lang et al., \*Archives of Physical Medicine and Rehabilitation\*, 2009; full text](#)). Therapist time is the limit, not the patient's day.

Devices that move the hand exist, but they are costly, clinic-bound or closed:

- Clinical robotic gloves such as Gloreha Sinfonia (now BTL R-TOUCH PRO) offer passive and active-assisted finger training, but they are tethered clinic systems ([Exoskeleton Report](#)).
- The NEOFECT Smart Glove costs about \$2,000 and tracks active movement for games; it does not move a hand that cannot move itself ([Recovery After Stroke](#)).
- Research tendon-driven gloves show the approach works. The Columbia hand orthosis for stroke uses a single forearm-mounted gearmotor with about 100 N peak tendon force and a dorsal tendon network to extend four fingers ([Park et al., arXiv 1802.06131; ROAM Lab](#)). Exo-Glove Poly II uses one actuator with antagonistic tendons and a 104 g polymer glove, but its actuation unit weighs about 1.14 kg and sits on a desk or wheelchair ([Kang et al., \*Soft Robotics\*, 2019](#)).

No open, garage-buildable, wearable design exists that a clinic, a research group or a family can build, inspect and adapt for daily repetitive finger movement at home.

A caution sits at the center of this problem. Robotic passive mobilization has been studied mainly for short-term effects on spasticity and limb perfusion ([Gobbo et al., "Hand Passive Mobilization Performed with Robotic Assistance: Acute Effects on Upper Limb Perfusion and Spasticity in Stroke Survivors," \*BioMed Research International\*, 2017; PMC5637828](#)). Whether passive repetitions alone improve hand function is not established. FlexHand is a passive-motion device (decided by Amish, 2026-09-25, FXH-DDR-001, D3); active or intent-triggered modes are a later option, and it makes no therapeutic claim.

## ■ Users and context

| User                                                                                                   | Need                                                                                          | Context                                                                                               |
|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Stroke survivor with moderate hand impairment and mild flexor tone (MAS 1 to 1+) (subacute or chronic) | Many gentle, repeatable finger movements a day without waiting for a therapist                | Home, seated with the forearm resting on a table or pillow; 30 to 60 min sessions, one to three a day |
| Care partner                                                                                           | Put the device on and take it off quickly and safely; know what to do if something goes wrong | Home                                                                                                  |
| Occupational or physical therapist                                                                     | Set range-of-motion, speed and force limits; see how many cycles were done                    | Clinic visit, then remote review                                                                      |
| Researcher or open hardware community                                                                  | A documented, reproducible platform to study dose and control strategies                      | University labs, makerspaces                                                                          |

## ■ Constraints

- Garage-buildable prototype, with a value-engineering target of USD 500 in parts (a control target, not a limit), using off-the-shelf motors and modules and 3D-printed parts.
- Wearable during a seated session: the hand side must be light and the actuators must sit on the forearm, not the hand.
- Fits a range of adult hands without custom molding.
- Safe with users who may have reduced sensation, spasticity and limited ability to remove the device themselves.
- Research and educational use only. FlexHand is not a medical device, has not been cleared or approved by any regulator, and must not be used to diagnose or treat any person outside a supervised research setting.

**Safety:** The device applies force to joints that may be spastic, contracted or insensate, and it carries a lithium-ion battery. Force limits, a physical stop, a tool-free tendon release and supervised use are part of the problem definition, not extras.

## ■ Out of scope

- Thumb actuation (the thumb is held in a passive abducted position in this concept).
- Wrist actuation.
- Diagnosis, assessment scoring or any therapeutic claim.
- Cloud services.

## ■ Open questions

- Which clinical partner to co-design with first (a stroke unit occupational therapy team, a community rehabilitation service, or a university rehabilitation lab)? Decided 2026-10-02 (FXH-DEC-001): a university rehabilitation lab attached to a stroke service, introduced through the OpenRatio network; first candidate to approach: Shirley Ryan AbilityLab in Chicago, or a university rehabilitation program nearer Irving. Decided on

2026-09-25 by Amish (FXH-DDR-001): passive motion only for this design, with active assist recorded as a later option (D3). The pitch is unchanged.

Decided on 2026-09-25 by Amish (FXH-DDR-002, N1): the first severity band is mild flexor tone (MAS 1 to 1+), which the 30 N design load covers according to the published finger stiffness check in FXH-CAL-001. Moderate and severe tone need roughly 80 to 150 N per finger and are left for a later design.